

SCENE-LEVEL TYPE-CONSISTENT RELABELLING FOR TRANSPARENT GLASS DETECTION

Heunseung Lim^{*†}, Jihyeon Ryu[†], Jinhoi Kim, Youngmin Son, Seongwoong Kim, and Gichul Yoo^{*}

SK intellix

ABSTRACT

GlassNICOL, a transparent-glass detection dataset, types each glass automatically from a depth-derived height in every view. Type is constant, so adjacent views should agree; the released training labels often disagree, while the hand-labelled test splits almost never do. We repair the labels with the structure that produced them. A scene is static, so every view of a glass is a view of one point on the table: we register views through the table plane, carry each box by its base point, chain the views of each glass, and give every box in a chain the chain’s most frequent type. Detectors trained on the relabelled set, five (accelerator, seed) cells under a sealed protocol, gain out of distribution in type accuracy and six-class average precision over raw labels and over an adjacent-frame vote; on the standard split six-class average precision does not improve and recall falls about two points. Only type ids change.

Index Terms— transparent object detection, label noise, multi-view consistency, relabelling, glass detection

1. INTRODUCTION

Transparent objects are expensive to label: depth sensors return little through glass and boundaries are weak, so ground truth comes from a proxy the sensor can see. ClearGrasp swaps each object for a spray-painted copy [1]; KeyPose labels keypoints in a few views and propagates them through known camera poses, using an opaque twin only for depth [2]; GlassNICOL photographs each tabletop scene in 25 views, records it again with the glasses capped to recover depth, and types each box by matching its depth-derived height to the nearest known glass [3]. KeyPose checks its geometry across views; nothing checks that the type assigned in one view agrees with the next.

We check. After registering adjacent training views and matching boxes, 18.1% of matched box pairs that are the same physical glass carry a different type; the hand-labelled test splits pass the same test at 0.0% and 0.3% (Section 2.1).

The structure that produced the labels can also repair them, and the scene says how. A GlassNICOL scene is static: the glasses stand on the table and only the head camera moves. Every view of a glass is therefore a view of one point of the table plane, its base, and a homography of that plane carries points of it exactly from one view to another. We register views pairwise through the table plane, carry each box by its base point, link boxes whose base points coincide, and merge the links into one chain per glass across the scene; each chain gives its most frequent type to every box in it (Fig. 1, Section 2). Our earlier baseline linked only adjacent frames by box overlap, and such chains break at a missed frame; registering through the table plane and carrying base points over two hops raises the median votes

per box from 5.5 to 10 while merging no two glasses of different type on the manual splits. 12.3% of type ids change and no box moves.

Contributions: (i) a measurement of type-label inconsistency in a released benchmark against its own manual test labels; (ii) a scene-level relabelling method that uses only the dataset’s own views, associates through the support plane without camera poses or depth, and changes type ids alone; (iii) evidence over five (accelerator, seed) cells on two accelerators under a protocol fixed before training (Section 3).

Relation to prior work. Repeat-factor sampling [4] and data pruning [5, 6] choose which images the network sees and take every label as given; relabelling changes what a label says. Label noise is usually found from model confidence [7, 8] or the training loss [9], or absorbed by robust training [10], in detectors too [11]; we locate it from the capture geometry, without a model. Tubelet methods smooth a detector’s own output at test time, by re-scoring [12, 13] or propagation [14] and, in one case, a majority over a pseudo-tracklet [15]; off-board auto-labelling turns detections over a sequence into training labels [16]. We vote on released labels, not on detections. Ground-plane homographies associate people across cameras by their feet [17]; we use the table plane the same way, across views of one moving camera. Label Fusion reprojects one 3D label into every frame [18]; GlassNICOL types each view separately, which is where the disagreement enters. KeyPose discards scans whose keypoints disagree [2]; we keep the frame and correct the label.

2. PROPOSED METHOD

2.1. Type labels disagree on the same object

GlassNICOL assigns type automatically (we call this step the labeller): the height of a depth cluster above the table plane, measured in the capped pass, is matched to the nearest of six known glass heights (5, 9, 11, 13, 17.5 and 22 cm) within 3 cm, all constants of the released labelling code [3]. Height is measured per view, but type is constant for the object, so one glass should carry one type across the 25 views of its scene. We register each of the 2,280 consecutive training frame pairs by background features, carry every box through the registration and match it to the box it overlaps in $f+1$ at intersection over union (IoU) above 0.5; pairs whose registration fails are matched unwarpd. Of the 2,775 matched pairs, 502 (18.1%) carry a different type (Fig. 2), 87% of them between glasses of neighbouring height.

The dataset authors likewise use manual test annotations to reduce noise from automatic labelling [3]. Those splits yield 0 flips in 93 matched pairs on the standard test, at most 1 in 316 (0.3%) on the out-of-distribution (OOD) test. The labeller also places two boxes on one glass in 51 training frames, 44 of 53 such pairs with different types, and none in the manual splits.

The misreads mostly join neighbouring heights and mostly run

[†]Equal contribution. ^{*}Corresponding authors: Heunseung Lim and Gichul Yoo.

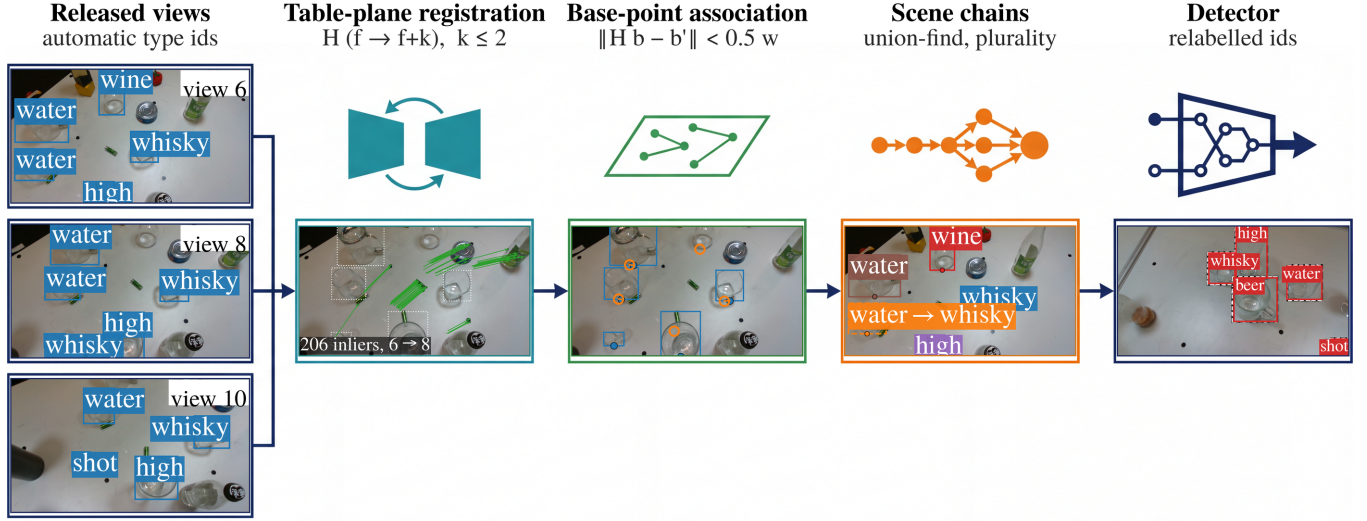


Fig. 1. Overview. Views of a static scene are registered through the table plane, each box is carried by its base point and linked to the box it lands on, union-find merges the links into one chain per glass, and the chain’s most frequent type replaces every type id in it. Photographs: views 6, 8 and 10 of one training scene with released boxes and types; inliers of the view 6 to 8 homography as displacement vectors; base points of view 6 carried into view 8 (open) and linked to its own (filled); view 6 with one colour per chain and a dashed box where the vote replaced the type; a scene-vote detector on a manual OOD test frame, manual boxes dashed, hand-picked where every manual box is detected with the correct type (Fig. 3 chooses its frames by rule). Overlays are real; stage symbols are schematic.

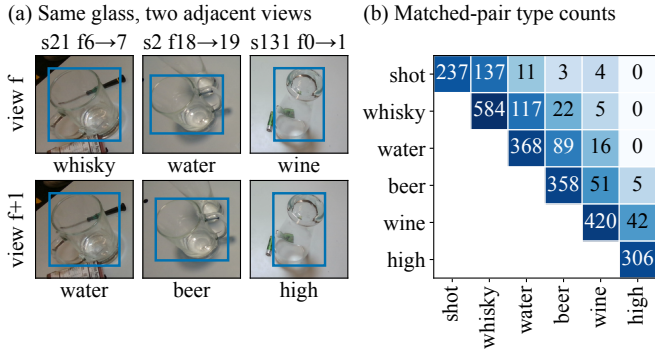


Fig. 2. Automatic type labels are not constant on one object. (a) The same glass in adjacent training views f and $f+1$, cropped from released frames with the released box and type. (b) Released types of the 2,775 box pairs matched across adjacent views after background registration, each unordered pair counted once; 502 (18.1%) disagree, 87% of them between neighbouring heights. Manual test labels flip 0/93 and 1/316.

downward. Treating the views of one glass on an adjacent-frame chain (the tracklets of Section 2.3) as repeated readings and estimating the labeller’s confusion by expectation–maximisation [19], a water glass is read as a shorter type in 18.5% of views and as a taller one in 4.7%, a beer glass 15.9% against 0.9%, a wine glass 11.0% against 0.8%, a highball 11.9%; only the shot glass, which has no shorter type, is misread upward, as whisky in 8.9%; the full estimated confusion is released with the code.

2.2. Association through the table plane

Frames f and $f+k$ of a scene, $k \leq 2$, are registered by background features: up to 1,500 ORB features [20] are extracted outside the label boxes, which are dilated $1.2\times$ and masked out, matched under

a 0.75 ratio test, and a homography is fitted by RANSAC (random sample consensus) [21] at a 3-pixel threshold and kept when at least 30 matches are inliers and they are at least half of the matches. The background is the table, so the homography is that plane’s and carries points lying on it exactly. We take a glass’s base point to be the bottom-centre of its box, an approximation the tolerance below absorbs; the rest of the box moves with parallax. The release annotates a base point per glass for the training split only, not for the hand-labelled splits where we check the association, so we read it off the box. Each box of f is carried into $f+k$ by its base point and linked, one-to-one in order of distance, to the box whose base point lies within 0.5 of the mean box width. Union-find merges links across the scene into chains, one glass seen in several views, under one constraint: a chain holds at most one box per frame, and a link that would violate it is refused (64 of the links proposed by the 4,097 registered pairs).

2.3. Vote, coverage and validity

Each chain takes its most frequent type; ties are left unchanged. Only the type id changes: no box moves, no image is added or removed, and roster and hyperparameters match the raw-label arm of Section 3. The association runs on all 95 training scenes, of which the detector roster below uses 70: 4,097 of 4,465 frame pairs register, and 4,336 boxes fall into 786 chains with a median of 10 views per box, 91% voted on by three views or more, 535 (12.3%) changing type and 140 in tied chains keeping their label. The adjacent-frame overlap vote of our earlier baseline, which registered 1,748 of 2,280 pairs by a partial-affine warp, gives 1,561 chains, 878 of them a single box, a median of 5.5 votes per box, and changes 364 boxes (8.4%). Ablating the two ingredients under the homography registration (post hoc; full table with the code) separates them: overlap linking gives a median of 8 votes over one hop and 9 over two, the base point over one hop 9 votes but 2 disagreements on the OOD split, and both together 10 votes at 12.3% of ids changed. Three of the four settings merge no

two glasses of different type on either manual split; among those three, ours votes with the most views. That baseline is the tracklet arm of Section 3. The validation split is left as released: checkpoint selection uses generic average precision (AP), so type ids never enter it.

Validity. Manual types are constant per object, so on the hand-labelled splits every within-chain type disagreement is a merged pair of glasses. The association above produces 0 of 167 disagreeing boxes on the standard split and 0 of 351 on the OOD split, with median chain lengths of 10 and 15 (163 of 167 and 346 of 351 boxes in chains of three or more). Hops and tolerance were chosen by this check, on a grid of two to four hops and four tolerances, before any scene-vote detector was trained, taking the loosest setting that disagrees nowhere; the zero above is therefore in-sample, and at the chosen tolerance three and four hops leave four disagreements on the standard split. The manual splits never enter training.

Scope. A vote over views removes disagreement that varies by view and inherits whatever is constant across views, such as the class-share difference of Section 4.

3. EXPERIMENTAL RESULTS

3.1. Setup

Data. The official GlassNICOL head-camera split [3]: 2,375 training images from 95 scenes, 125 validation images, and hand-labelled standard and OOD tests (66 and 75 images, 3 scenes each). The roster, fixed beforehand, uses 70 scenes (1,750 images).

Detector and regime. YOLO11n [22] (2.59M parameters [23]), 640 × 640 inputs, batch 64, AdamW [24] at $\text{lr}_0 = 10^{-3}$, cosine decay over 300 epochs. All arms train under the same repeated-exposure regime: the 1,538 training images whose transparent and chalk-coated captures register by the same background matching appear twice, 3,288 slots and 15,600 optimiser updates, and the arms differ only in training type ids: raw, tracklet vote, scene vote. Seeds 1337, 3407 and 4567 run on accelerator A (an H200 GPU) and seeds 1337 and 5678 on accelerator B (an RTX PRO 6000 GPU); a cell is one (accelerator, seed) set of the three arms, an arm being one label variant on the shared roster. On A the raw and scene-vote runs of a cell share a GPU and the tracklet run used another; on B all share one.

Scoring. The checkpoint with the best validation generic AP is scored at confidence floor 0.001 with non-maximum suppression (NMS) at IoU 0.7: COCO (Common Objects in Context) box AP over IoU 0.50–0.95 [25]; generic AP collapses the six types to one. Within each image, GT boxes in ascending annotation-id order greedily take the unused prediction with maximum IoU (≥ 0.5 , confidence ≥ 0.25), irrespective of type. Pairwise type accuracy is the correctly typed fraction on boxes localised by both compared arms; recall is the localised fraction per arm.

Existing methods compared. Two standard answers to label noise are trained as arms under the same protocol. Label smoothing [26] keeps the released ids and smooths the classification target of the assigned anchors by 0.1. Detector-confidence relabelling replaces a type id when a detector disagrees with confidence at least 0.5, judged by a model trained on the scene-disjoint half of the roster that excludes that box, so the judge sees half the scenes at a quarter of the updates; it is the confident-learning idea [7, 27] at one global threshold, flipping rather than pruning, and changes 22.5% of the ids on that roster against our 11.7%.

Protocol. The comparison was preregistered before any scene-vote detector was trained: the three arms, the five cells, the checkpoint

rule, the four metrics above, and four predictions, a positive mean type-accuracy gain over the raw labels on both splits (P1), an OOD gain exceeding the tracklet vote’s (P2), recall and generic AP within two points (P3), and a positive OOD gain in at least 4 of 5 cells (P4). The sealed protocol, every evaluation receipt and the code are at <https://github.com/HeunSeungLim/glassnicol-scene-relabel>. Every contrast is read as a mean over cells with the count of positive cells beside it [28].

3.2. What the labels change

Table 1 pools the cells; the per-cell table with its intervals is released with the code. Two contrasts hold across cells. Out of distribution, six-class AP rises by +7.1 points over the raw labels in 5 of 5 cells, +6.4 on accelerator A and +8.0 on B (across-cell 95% t -interval [+1.8, +12.3]) and by +6.4 over the tracklet vote ([+0.6, +12.2]), which the tracklet vote itself did not move (+0.7). On the standard split localisation recall falls in every cell, by 2.0 points on average ([−3.7, −0.3]). The type-accuracy gains are larger but less certain: out of distribution the scene vote gains +9.2 points over the raw labels, positive in 4 of 5 cells, with per-cell paired bootstrap intervals over 2,000 draws that resample test images above zero in 3 cells, reaching it in 1 and below it in 1; resampling whole scenes instead leaves 8 of the 10 per-cell intervals over both splits including zero, and the across-cell interval is [−5.7, +24.1]; over the tracklet vote it gains +4.0 (4 of 5). On the standard split type accuracy changes by +0.8 over the raw labels (3 of 5 cells positive); on the shared denominator of Table 1 it is 84.1 against 84.2, so the two box sets disagree in sign on a draw and −1.8 over the tracklet vote (1), while six-class and generic AP change by −1.9 and −1.7 (2 and 1 of 5 positive).

Preregistered predictions. P1 held: the mean type-accuracy gain over the raw labels is positive on both splits, +0.8 and +9.2. P2 held: out of distribution the gain exceeds the tracklet vote’s +5.0 (pairwise contrasts use different common-box sets and need not add). P4 held: 4 of 5 OOD cells are positive. P3 did not: standard-split recall falls 2.0 points, just past the bound, and generic AP 1.7; out of distribution both stay within it. Scoring the last epoch instead (post hoc) gives type accuracy −0.6 on the standard split, so P1 holds only under the preregistered rule, and +11.3 out of distribution; recall moves +0.5 and −1.7, so the OOD gain holds under both rules while the standard-split recall loss does not. Under that rule the tracklet vote also gains out of distribution, six-class AP +7.0 in 5 of 5 cells, but the scene vote still leads it by +4.1 points in 5 of 5 ([−0.2, +8.3], against [+0.6, +12.2] under the preregistered rule): the lead survives the rule change, but its interval now includes zero.

3.3. Comparison with existing methods

Table 1 sets the two standard answers to label noise, and our earlier vote, beside ours; Fig. 3 shows the same five detectors on manual test frames. Label smoothing leaves the detector where the released labels leave it (−0.5 out of distribution): absorbing the noise does not recover a wrong type id. Detector confidence moves the labels the most and the scores furthest the wrong way, −15.5 out of distribution and above the released labels in 0 of 5 cells. It rewrites 22.5% of the ids against our 11.7%, and 81.8% of those differ from the type the geometry votes for, of which 48.2 points are forced by the edit counts alone: a box it moves and we leave alone differs by construction. Under the preregistered rule the scene vote alone gains out of distribution in every cell, +7.1 points in 5 of 5, against +0.7 in 2 for the adjacent-frame vote. At threshold 0.8 (post hoc, accelerator A) it edits 12.8% of the ids, about as many as our vote, and still falls

Table 1. All five training-label variants on the official test splits (manual labels), mean over the 5 (accelerator, seed) cells; Δ is the change from the released labels before rounding, with the number of cells above them. Six: six-class AP; Gen: generic AP, the six types collapsed to one; Rec: localisation recall; Type: type accuracy. Every row trains the same detector on the same roster, budget and seeds, three cells on accelerator A and two on B, and the rows differ only in the training type ids, or, for label smoothing, in the classification target. Type is scored on the boxes all five arms localise in a cell (121–135 of 167 on the standard split, 222–273 of 351 out of distribution), one denominator for the five columns; the preregistered predictions of Section 3 use pairwise common sets instead and do not have to agree with it. The two external rows are post hoc, added after the sealed protocol. Best per column bold, second blue.

Training labels	Standard					OOD				
	Six	Δ	Gen	Rec	Type	Six	Δ	Gen	Rec	Type
Released labels	65.4	—	84.2	90.8	84.2	53.7	—	66.6	79.7	76.0
Label smoothing	68.7	+3.3 (3/5)	84.2	91.3	84.7	53.1	−0.5 (3/5)	65.6	79.9	76.4
Detector confidence	62.2	−3.1 (1/5)	83.9	88.3	81.6	38.2	−15.5 (0/5)	65.4	79.0	62.9
Adjacent-frame vote (earlier)	65.6	+0.2 (3/5)	85.7	89.0	86.1	54.3	+0.7 (2/5)	67.1	80.1	81.3
Scene vote (ours)	63.5	−1.9 (2/5)	82.4	88.7	84.1	60.7	+7.1 (5/5)	66.3	78.4	84.8

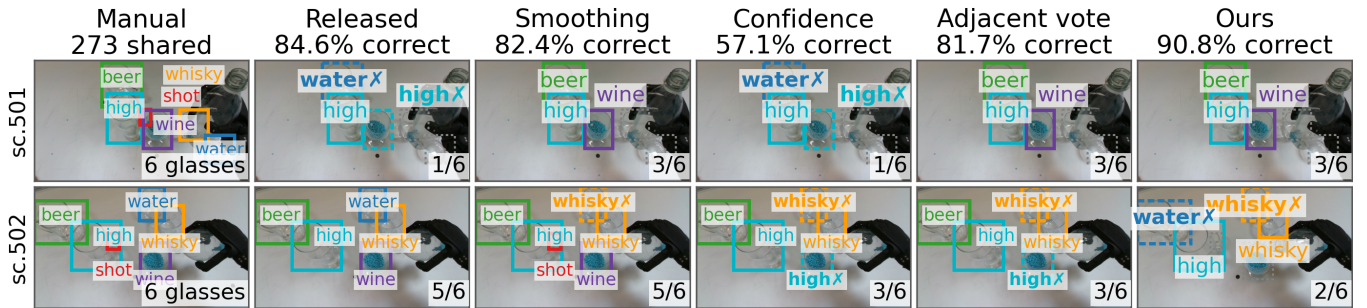


Fig. 3. The five training-label arms on OOD test frames, accelerator A seed 1337, from the frozen predictions behind Table 1. Columns: the manual annotation, then the detectors trained on released labels, label smoothing, detector-confidence relabelling, the adjacent-frame vote and ours. Boxes sit at the manual location, coloured by the predicted type, solid if right, dashed with a cross if wrong, grey dotted if the arm localises nothing there. Headings score all arms on the 273 boxes every arm localises, one shared denominator; a panel count is correctly typed boxes of the frame’s manual boxes, so one no arm finds counts against all. Row 1 is a frame where the released-label arm gets the most boxes wrong, mistyped or missed (four tie; first by file name); row 2 is our worst frame, the unique maximum of boxes we get wrong, again mistyped or missed, that it gets right.

11.2 points out of distribution in all 3 cells while beating us on the standard split: fewer edits soften the loss without turning it into a gain. Smoothing is the better choice on the standard split, ahead of us in every cell, so the case for relabelling is the out-of-distribution one. The last epoch keeps the order: +0.7 for smoothing, −4.2 for detector confidence, +7.0 for the tracklet vote and +11.1 for ours in 5 of 5.

4. DISCUSSION AND LIMITATIONS

The vote does not bring class shares closer to the tests: the shot:whisky ratio is 0.58 in raw labels and 0.51 after it, against 0.79 and 1.83 in the tests; decoding the chains under the estimated confusion moves it no closer, and separating the labeller’s contribution from the staging’s needs the per-box height the release omits.

Six-class AP is the mean over the six types, so the standard-split −1.9 is a reallocation that nets out rather than a uniform decline: shot gains +12.4 and beer +5.7 against −14.6 for whisky and −12.3 for water, while type accuracy on that split rises +0.8. Out of distribution the reallocation runs one way, +17.6 on water and +17.2 on wine, with no type below −0.7. The edits are short and mostly upward: 77.8% of the 535 changed ids move a single step on the height ladder and 68.4% move up. The shortest type can only move up and the tallest only down; over the 395 edits that start at neither, 62.5% still move up, undoing the downward bias the labeller’s estimated confusion

shows.

The association is only as good as the table: frames without enough background features stay unlinked, and the 51 frames with two boxes on one glass are counted twice.

Seed 4567 on A loses type accuracy on both splits, by eight and ten points, while every other cell gains six or more out of distribution, and the type-accuracy mean sits inside the seed spread, so six-class AP carries the claim; the tracklet contrast rerun on B for seed 1337 moved by 10.9/14.3 points, a caution for single-cell results. Checkpoint selection by validation generic AP ($r^2 = 0.04$ with test six-class AP) is shared across arms; only labels change.

5. CONCLUSION

A released benchmark can disagree with itself, measurably and without a model: GlassNICOL’s automatic types differ on the same glass in 18.1% of adjacent-view pairs, where its hand labels almost never do. The geometry that produced them repairs them: a homography of the static table plane carries base points between views, union-find chains them, and a plurality settles the type. Only 12.3% of ids move, no box does. The repair pays out of distribution, where under the preregistered rule ours alone gains in every cell, +7.1 six-class AP, while smoothing leaves the detector where it was and a detector-written rewrite moves it −15.5. Where a pipeline leaves a redundancy unused, that redundancy is a repair.

6. REFERENCES

- [1] Shreeyak S. Sajjan, Matthew Moore, Mike Pan, Ganesh Nagaraja, Johnny Lee, Andy Zeng, and Shuran Song, “ClearGrasp: 3d shape estimation of transparent objects for manipulation,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 3634–3642.
- [2] Xingyu Liu, Rico Jonschkowski, Anelia Angelova, and Kurt Konolige, “KeyPose: Multi-view 3d labeling and keypoint estimation for transparent objects,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 11599–11607.
- [3] Lukáš Gajdošech, Hassan Ali, Jan-Gerrit Habekost, Martin Madaras, Matthias Kerzel, and Stefan Wermter, “Shaken, not stirred: A novel dataset for visual understanding of glasses in human-robot bartending tasks,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2025, pp. 20516–20523.
- [4] Agrim Gupta, Piotr Dollár, and Ross Girshick, “LVIS: A dataset for large vocabulary instance segmentation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 5356–5364.
- [5] Mansheej Paul, Surya Ganguli, and Gintare Karolina Dziugaite, “Deep learning on a data diet: Finding important examples early in training,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021.
- [6] Ben Sorscher, Robert Geirhos, Shashank Shekhar, Surya Ganguli, and Ari S. Morcos, “Beyond neural scaling laws: Beating power law scaling via data pruning,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [7] Curtis G. Northcutt, Lu Jiang, and Isaac L. Chuang, “Confident learning: Estimating uncertainty in dataset labels,” *J. Artif. Intell. Res.*, vol. 70, pp. 1373–1411, 2021.
- [8] Curtis G. Northcutt, Anish Athalye, and Jonas Mueller, “Pervasive label errors in test sets destabilize machine learning benchmarks,” in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, 2021.
- [9] Marius Schubert, Tobias Riedlinger, Karsten Kahl, Daniel Kröll, Sebastian Schöner, Siniša Šegvić, and Matthias Rottmann, “Identifying label errors in object detection datasets by loss inspection,” in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2024, pp. 4582–4591.
- [10] Hwanjun Song, Minseok Kim, Dongmin Park, Yooju Shin, and Jae-Gil Lee, “Learning from noisy labels with deep neural networks: A survey,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8135–8153, 2023.
- [11] Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, and Yixuan Yuan, “Towards robust adaptive object detection under noisy annotations,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 14187–14196.
- [12] Kai Kang, Wanli Ouyang, Hongsheng Li, and Xiaogang Wang, “Object detection from video tubelets with convolutional neural networks,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 817–825.
- [13] Wei Han, Pooya Khorrami, Tom Le Paine, Prajit Ramachandran, Mohammad Babaeizadeh, Honghui Shi, Jianan Li, Shuicheng Yan, and Thomas S. Huang, “Seq-NMS for video object detection,” *arXiv preprint arXiv:1602.08465*, 2016.
- [14] Subarna Tripathi, Serge Belongie, Youngbae Hwang, and Truong Nguyen, “Detecting temporally consistent objects in videos through object class label propagation,” in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, 2016, pp. 1–9.
- [15] Yang Chen, Changsoo S. Jeong, Deepak Khosla, Kyungnam Kim, Shinko Y. Cheng, Lei Zhang, and Alexander L. Honda, “Object recognition consistency improvement using a pseudo-tracklet approach,” U.S. Patent 9,342,759 B1, HRL Laboratories, May 2016.
- [16] Charles R. Qi, Yin Zhou, Mahyar Najibi, Pei Sun, Khoa Vo, Boyang Deng, and Dragomir Anguelov, “Offboard 3D object detection from point cloud sequences,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 6134–6144.
- [17] Saad M. Khan and Mubarak Shah, “A multiview approach to tracking people in crowded scenes using a planar homography constraint,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2006, vol. 3954 of *LNCS*, pp. 133–146.
- [18] Pat Marion, Peter R. Florence, Lucas Manuelli, and Russ Tedrake, “Label Fusion: A pipeline for generating ground truth labels for real RGBD data of cluttered scenes,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2018, pp. 3235–3242.
- [19] A. P. Dawid and A. M. Skene, “Maximum likelihood estimation of observer error-rates using the EM algorithm,” *J. R. Stat. Soc. C (Appl. Stat.)*, vol. 28, no. 1, pp. 20–28, 1979.
- [20] Ethan Rublee, Vincent Rabaud, Kurt Konolige, and Gary Bradski, “ORB: An efficient alternative to SIFT or SURF,” in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2011, pp. 2564–2571.
- [21] Martin A. Fischler and Robert C. Bolles, “Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography,” *Commun. ACM*, vol. 24, no. 6, pp. 381–395, 1981.
- [22] Glenn Jocher and Jing Qiu, “Ultralytics YOLO11,” 2024, Software, <https://github.com/ultralytics/ultralytics>.
- [23] Ranjan Sapkota, Marco Flores-Calero, Rizwan Qureshi, Chetan Badgujar, Upesh Nepal, Alwin Poulouse, Peter Zeno, Uday Bhanu Prakash Vaddevolu, Sheheryar Khan, Maged Shoman, Hong Yan, and Manoj Karkee, “YOLO advances to its genesis: A decadal and comprehensive review of the you only look once (YOLO) series,” *Artif. Intell. Rev.*, vol. 58, no. 9, 2025, art. no. 274.
- [24] Ilya Loshchilov and Frank Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2019.
- [25] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick, “Microsoft COCO: Common objects in context,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 740–755.
- [26] Rafael Müller, Simon Kornblith, and Geoffrey Hinton, “When does label smoothing help?,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [27] Krystian Chachula, Bartłomiej Olber, Jakub Łyskawa, Adam Popowicz, and Krystian Radlak, “Combating noisy labels in object detection datasets,” *Machine Learning*, vol. 115, 2026, art. no. 78.
- [28] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent, “Accounting for variance in machine learning benchmarks,” in *Proc. Mach. Learn. Syst. (MLSys)*, 2021, vol. 3, pp. 747–769.